

```

%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;
scuts_path = 'e:\SHARE\FE\Data\analysis\07_fit_with_multicut_allJ\sel_cuts';
Cutfitfile = fullfile(scuts_path, 'GP_FFCutsFits.mat');

if ~exist('GP_FFCutsFits', 'var')
    ld = load(Cutfitfile);
    GP_FFCutsFits = ld.fit_src_struct;
end

```

```

fn = fieldnames(GP_FFCutsFits);
nf = 11

```

```

nf =
11

```

```

%for i = 11:numel(fn)
    %selectCuts = fn{nf};
    selectCuts = 'Ei400_off110GPmaxFF'

```

```

selectCuts =
'Ei400_off110GPmaxFF'

```

```

if contains(selectCuts, 'msk')
    error('Can not fit field: %s', selectCuts);
end
if contains(selectCuts, 'all_fit_par')
    error('Can not fit field: %s', selectCuts);
end

fprintf('Processing cuts: %s \n', selectCuts);

```

```

Processing cuts: Ei400_off110GPmaxFF

```

```

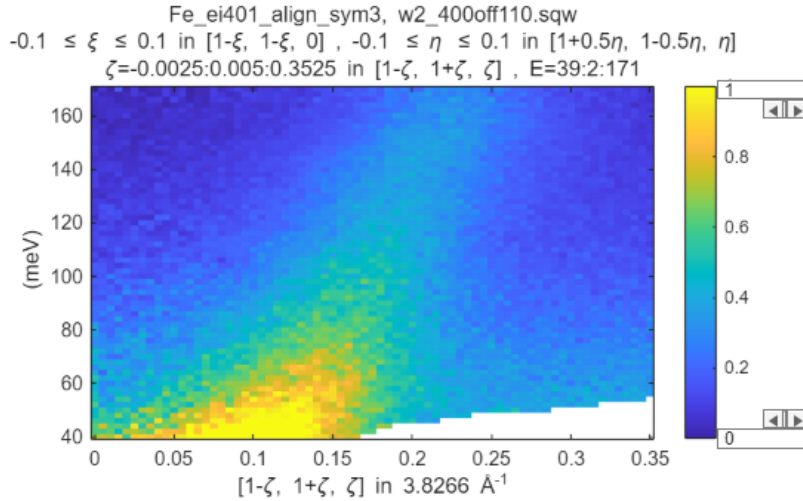
%selectCuts= 'Ei400_off200GPmaxFF';
cuts_list = GP_FFCutsFits.(selectCuts);
if ~iscell(cuts_list)
    cuts_list = {cuts_list};
end
if contains(selectCuts, 'Ei200')
    mi = maps_instrument(200, 600, 'S');
end
if contains(selectCuts, 'Ei400')
    mi = maps_instrument(401, 600, 'S');
end
if contains(selectCuts, 'Ei800')
    mi = maps_instrument(786, 600, 'S');
end
sample=IX_sample(true, [1,0,0], [0,1,0], 'cuboid', [0.04,0.03,0.02]);
for j=1:numel(cuts_list)

```

```

cuts_list{j} = cuts_list{j}.set_instrument(mi);
cuts_list{j} = cuts_list{j}.set_sample(sample);
plot(cuts_list{j}); lz 0 1; keep_figure;
end

```



```

dir_name = "GP";

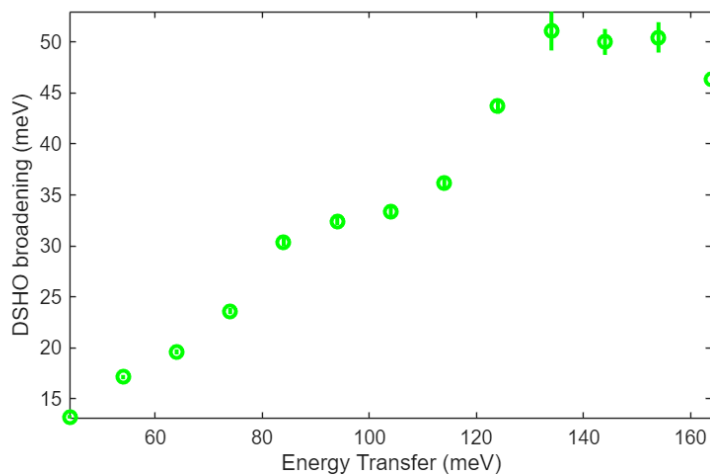
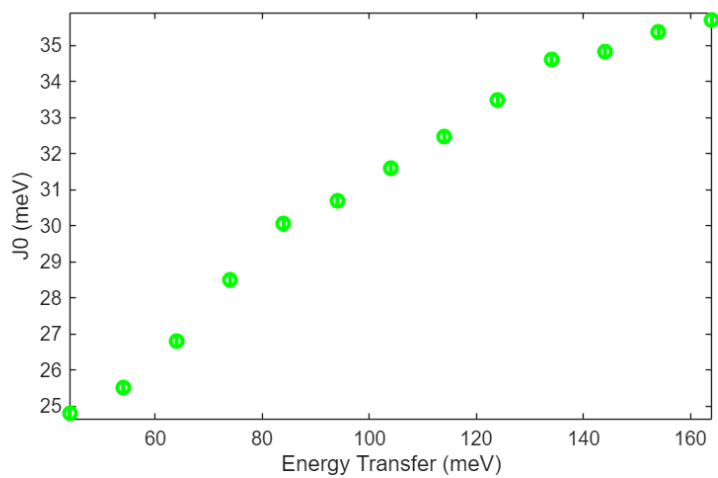
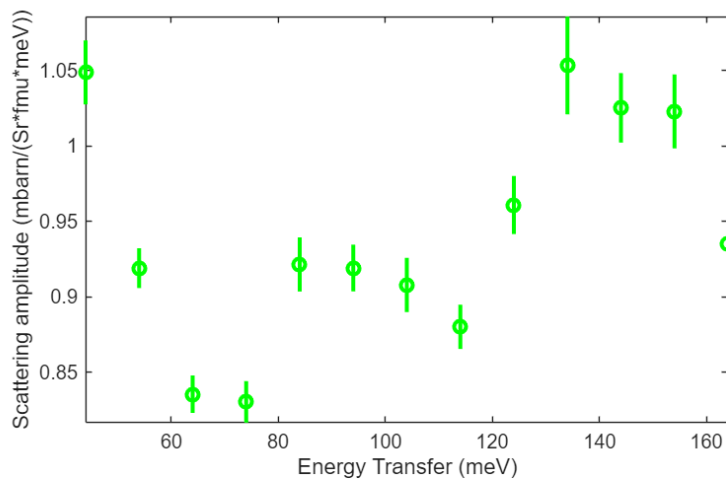
dE_step = 2; %original energy transfer step data were binned to. No point in
going finer
half_dE = 5; % half width of data binning

en_mm = cuts_list{1}.data.img_range(:,4);
en_mm(1) = max(30,en_mm(1));
%en_mm(1) = 100;
%en_mm(2) = 170;
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);

cut_dimensions = 1;
GP_FFCutsFits.(selectCuts) = cuts_list;

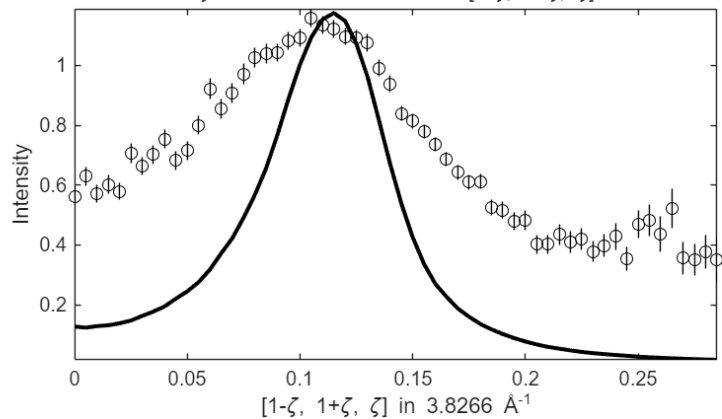
GP_FFCutsFits = fit_multicutsFF_along_direction(...
    GP_FFCutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);

```



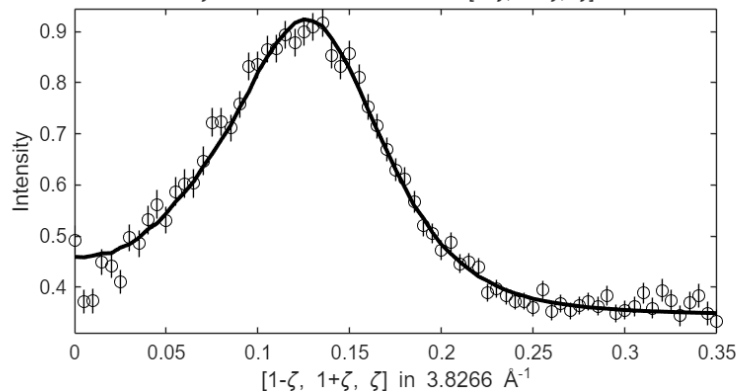
 **** En = 44±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $39 \leq E$
 $\zeta = -0.0025:0.005:0.3525$ in $[1-\zeta, 1+\zeta, \zeta]$

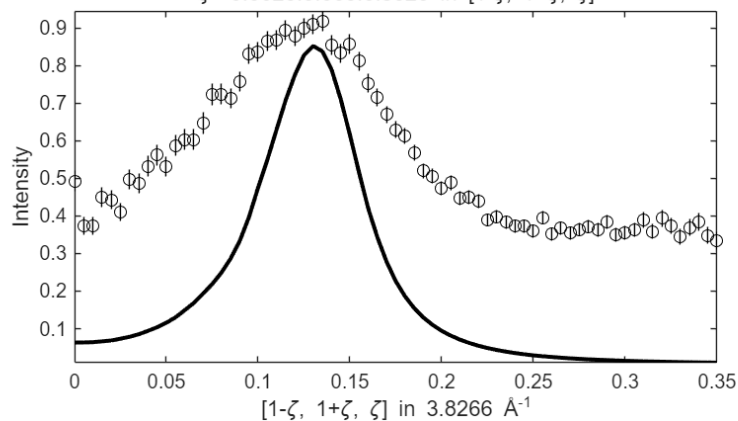


**** En = 54±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw
 $S=0.92 \pm 0.013$; $J_0 = 26 \pm 0.2$; $\gamma = 17.2248 \pm 0.211586$
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $49 \leq E$
 $\zeta = -0.0025:0.005:0.3525$ in $[1-\zeta, 1+\zeta, \zeta]$

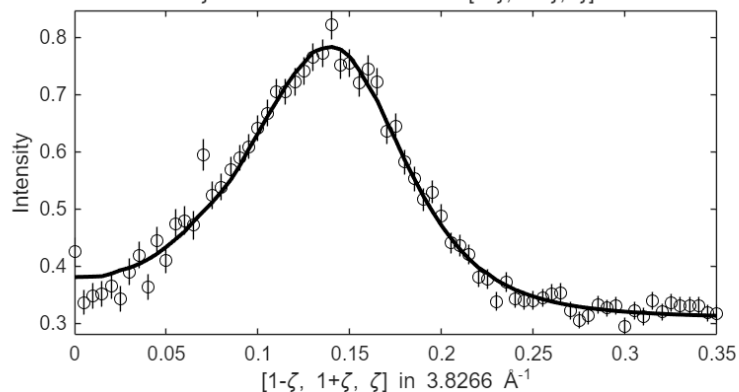


Fe_ei401_align_sym3, w2_400off110.sqw
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $49 \leq E$
 $\zeta = -0.0025:0.005:0.3525$ in $[1-\zeta, 1+\zeta, \zeta]$

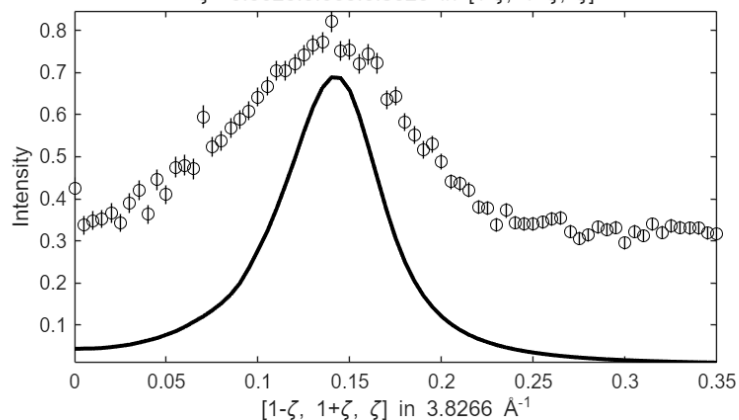


**** En = 64±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw
 $S=0.84\pm0.012$; $J_0=27\pm0.21$; $\gamma=19.6341\pm0.248602$
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $59 \leq E$
 $\zeta=-0.0025:0.005:0.3525$ in $[1-\zeta, 1+\zeta, \zeta]$

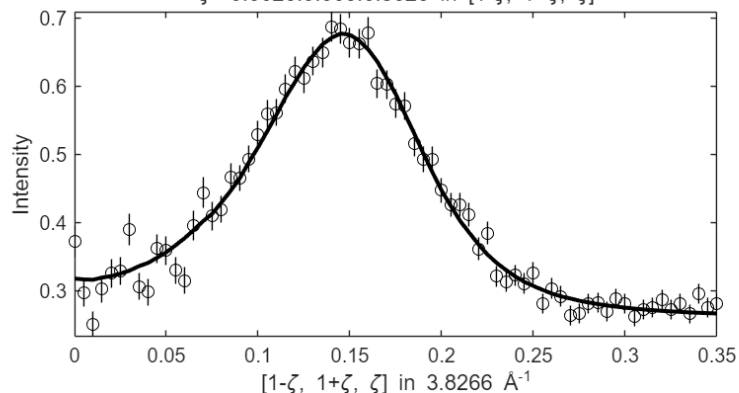


Fe_ei401_align_sym3, w2_400off110.sqw
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $59 \leq E$
 $\zeta=-0.0025:0.005:0.3525$ in $[1-\zeta, 1+\zeta, \zeta]$

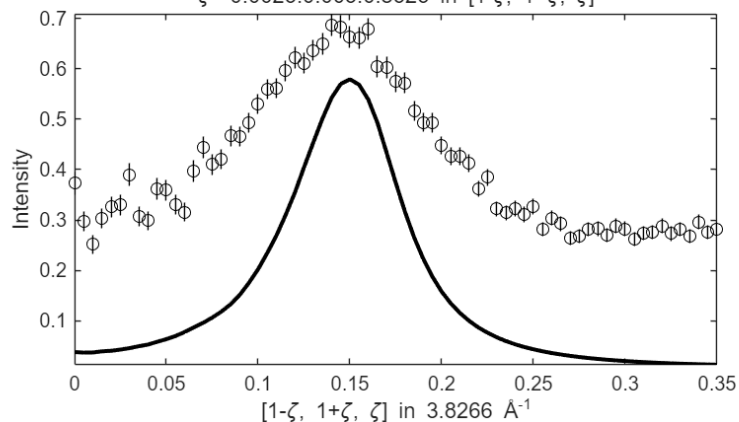


 **** En = 74±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw
 $S=0.83\pm0.049$; $J_0=29\pm0.35$; $\gamma=23.79\pm2.32805$
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $69 \leq E$
 $\zeta=-0.0025:0.005:0.3525$ in $[1-\zeta, 1+\zeta, \zeta]$

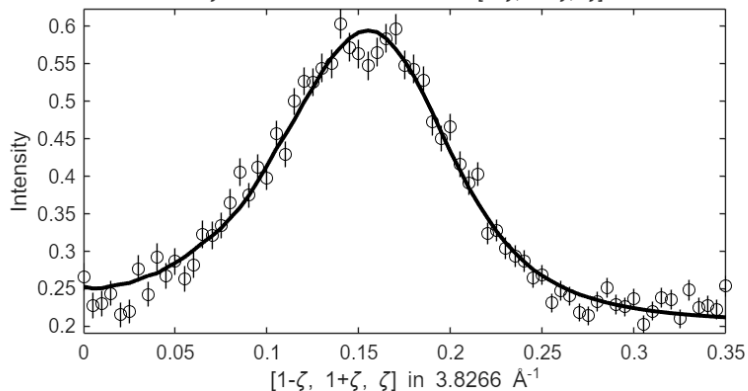


Fe_ei401_align_sym3, w2_400off110.sqw
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $69 \leq E$
 $\zeta = -0.0025:0.005:0.3525$ in $[1-\zeta, 1+\zeta, \zeta]$

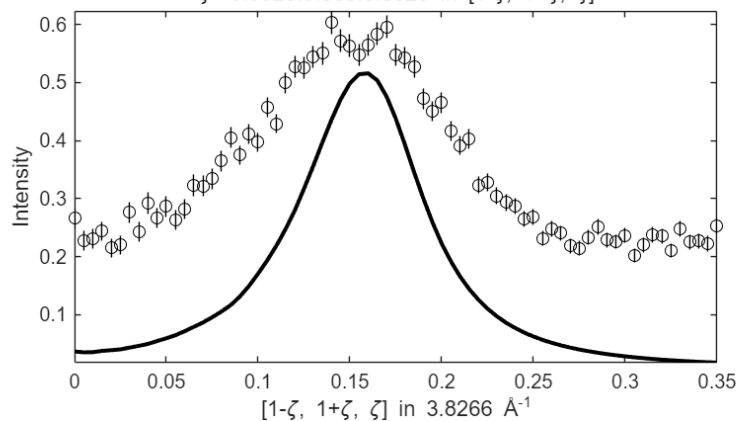


**** En = 84±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw
 $S=0.92 \pm 0.062$; $J_0 = 30 \pm 0.42$; $\gamma = 30.4487 \pm 3.07$
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $79 \leq E$
 $\zeta = -0.0025:0.005:0.3525$ in $[1-\zeta, 1+\zeta, \zeta]$

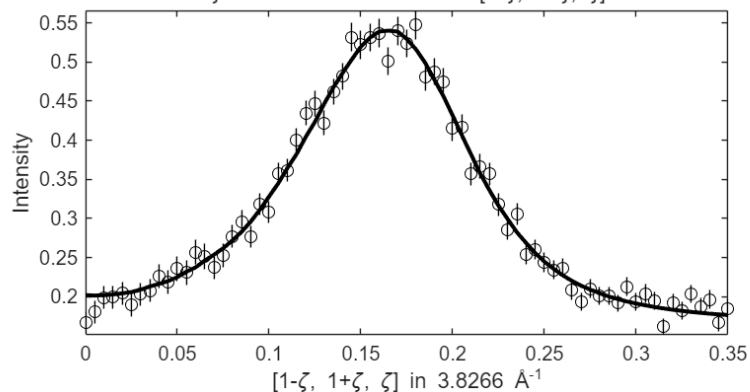


Fe_ei401_align_sym3, w2_400off110.sqw
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $79 \leq E$
 $\zeta = -0.0025:0.005:0.3525$ in $[1-\zeta, 1+\zeta, \zeta]$

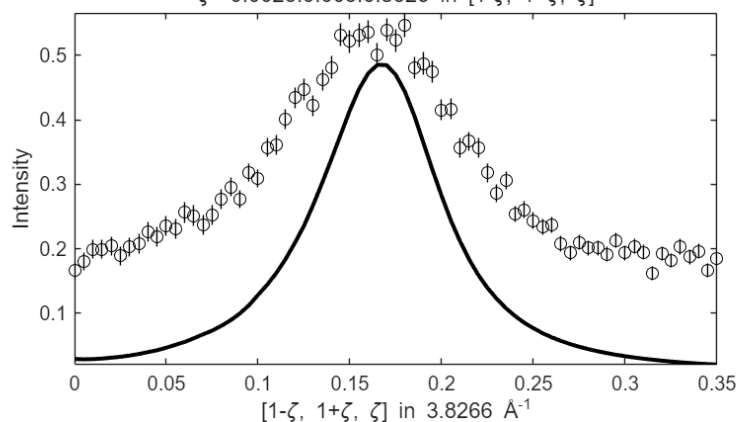


**** En = 94±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw
 $S=0.92\pm0.015$; $J_0=31\pm0.17$; $\gamma=32.2042\pm0.365846$
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $89 \leq E$
 $\zeta=-0.0025:0.005:0.3525$ in $[1-\zeta, 1+\zeta, \zeta]$

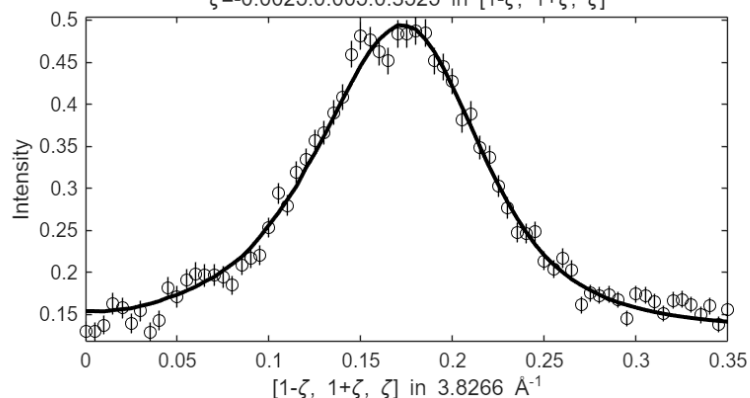


Fe_ei401_align_sym3, w2_400off110.sqw
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $89 \leq E$
 $\zeta=-0.0025:0.005:0.3525$ in $[1-\zeta, 1+\zeta, \zeta]$

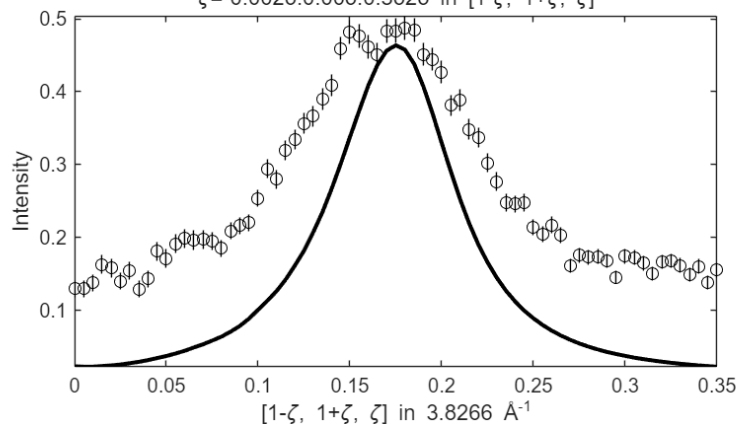


 **** En = 104±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw
 $S=0.91\pm0.018$; $J_0=32\pm0.17$; $\gamma=33.3071\pm0.486469$
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $99 \leq E$:
 $\zeta=-0.0025:0.005:0.3525$ in $[1-\zeta, 1+\zeta, \zeta]$

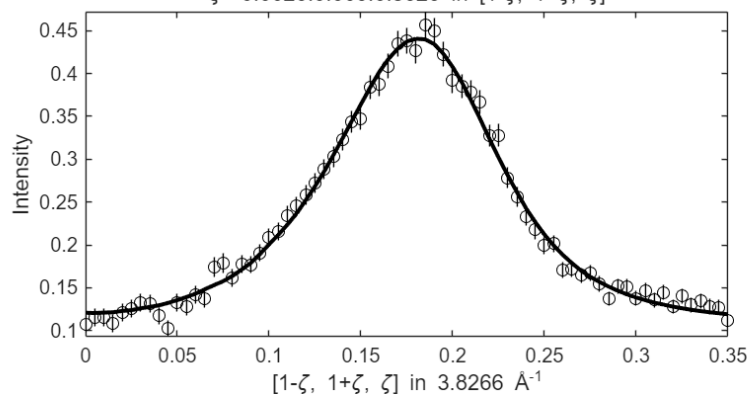


Fe_ei401_align_sym3, w2_400off110.sqw
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $99 \leq E$:
 $\zeta = -0.0025:0.005:0.3525$ in $[1-\zeta, 1+\zeta, \zeta]$

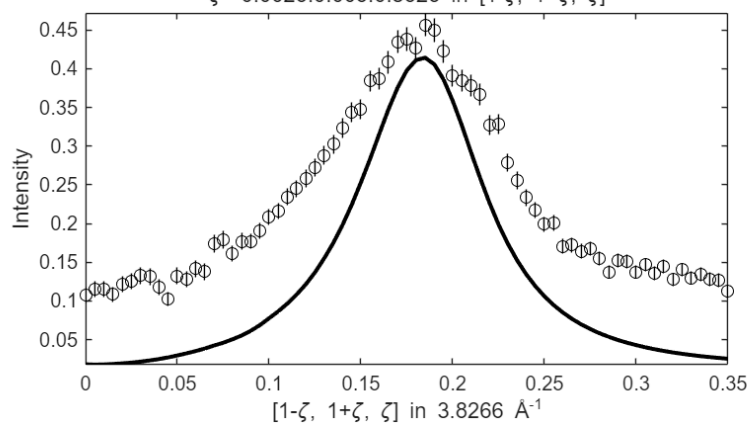


**** En = 114±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw
 $S=0.88 \pm 0.015$; $J_0 = 32 \pm 0.12$; $\gamma = 36.1641 \pm 0.46893$
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $109 \leq E$
 $\zeta = -0.0025:0.005:0.3525$ in $[1-\zeta, 1+\zeta, \zeta]$

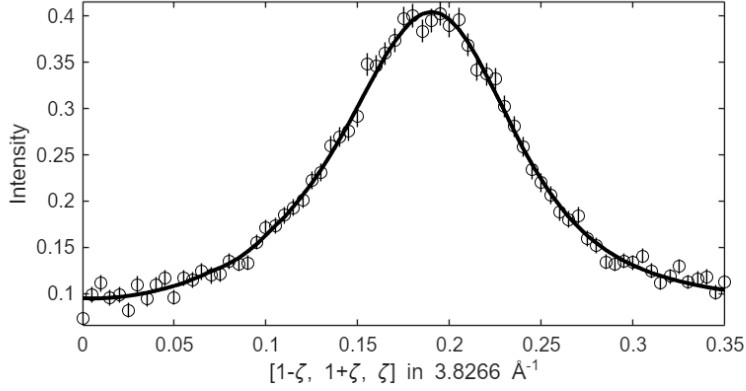


Fe_ei401_align_sym3, w2_400off110.sqw
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $109 \leq E$
 $\zeta = -0.0025:0.005:0.3525$ in $[1-\zeta, 1+\zeta, \zeta]$

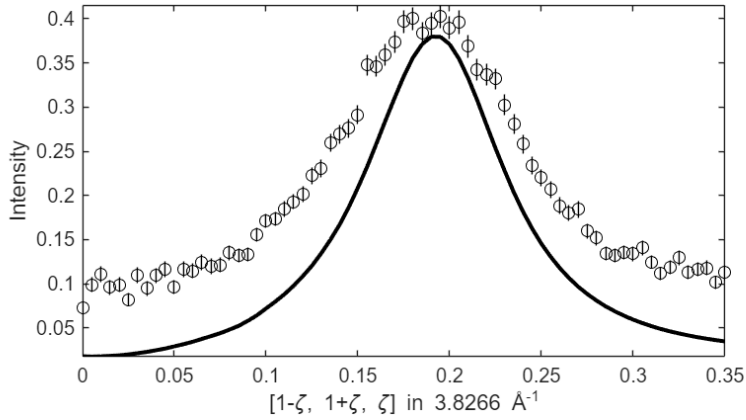


**** En = 124±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw
 $S=0.96\pm0.035$; $J_0=33\pm0.21$; $\gamma=43.7466\pm1.99395$
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $119 \leq E$
 $\zeta=-0.0025:0.005:0.3525$ in $[1-\zeta, 1+\zeta, \zeta]$

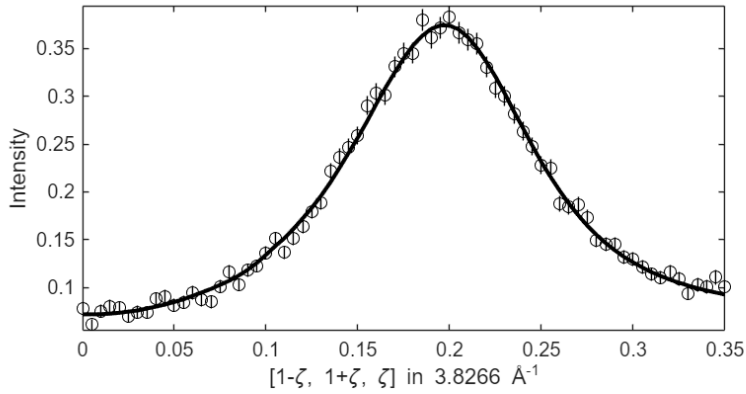


Fe_ei401_align_sym3, w2_400off110.sqw
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $119 \leq E$
 $\zeta=-0.0025:0.005:0.3525$ in $[1-\zeta, 1+\zeta, \zeta]$

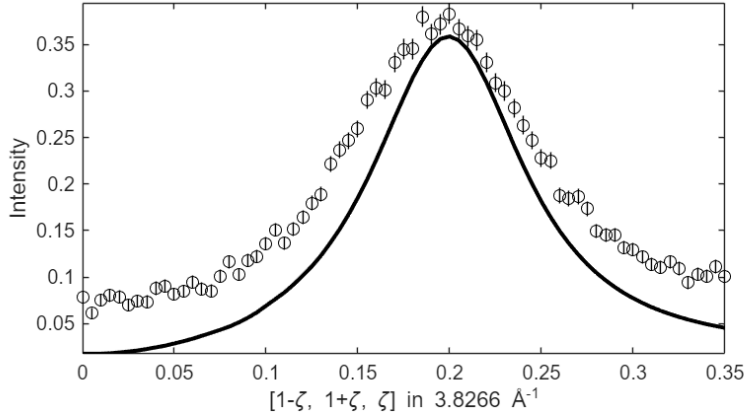


**** En = 134±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw
 $S=1.1\pm0.028$; $J_0=35\pm0.14$; $\gamma=51.0485\pm1.40596$
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $129 \leq E$
 $\zeta=-0.0025:0.005:0.3525$ in $[1-\zeta, 1+\zeta, \zeta]$

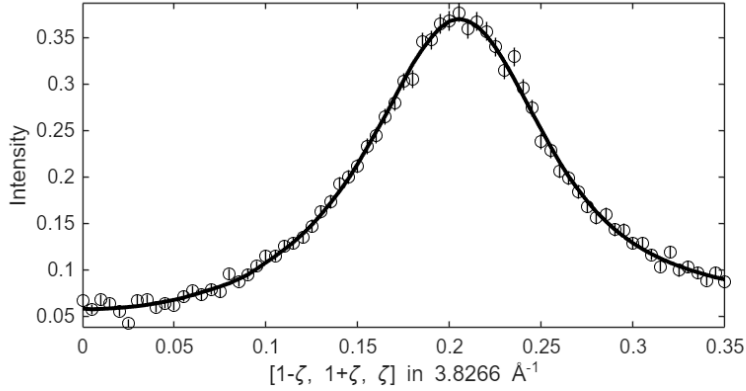


Fe_ei401_align_sym3, w2_400off110.sqw
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $129 \leq E$
 $\zeta = -0.0025:0.005:0.3525$ in $[1-\zeta, 1+\zeta, \zeta]$

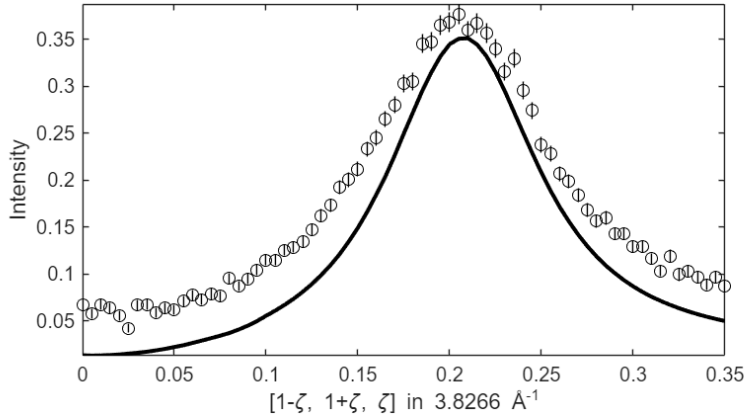


**** En = 144±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw
 $S = 1 \pm 0.023$; $J_0 = 35 \pm 0.13$; $\gamma = 50.0243 \pm 1.27655$
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $139 \leq E$
 $\zeta = -0.0025:0.005:0.3525$ in $[1-\zeta, 1+\zeta, \zeta]$

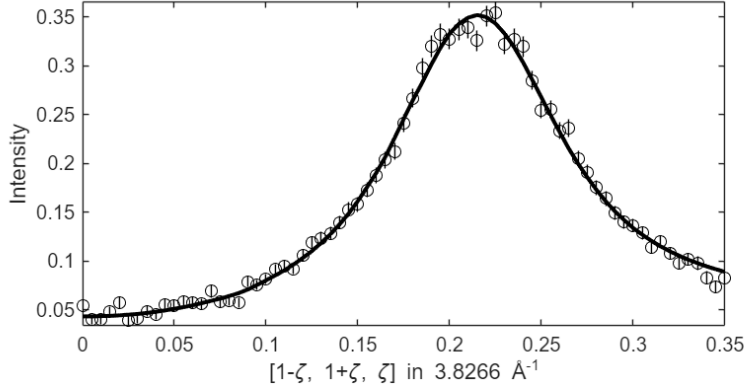


Fe_ei401_align_sym3, w2_400off110.sqw
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $139 \leq E$
 $\zeta = -0.0025:0.005:0.3525$ in $[1-\zeta, 1+\zeta, \zeta]$

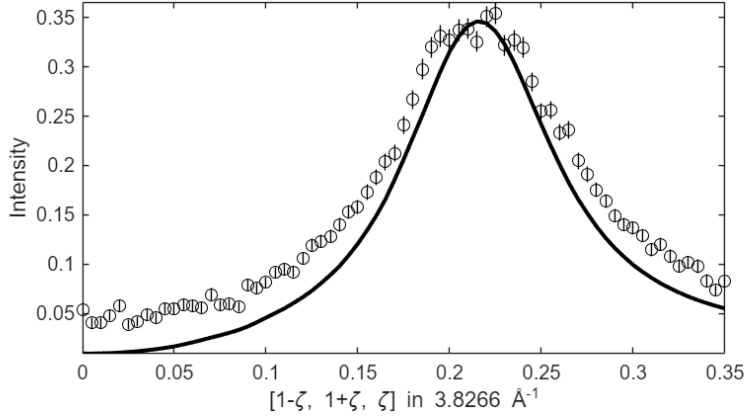


**** En = 154±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw
 $S = 1 \pm 0.025$; $J_0 = 35 \pm 0.15$; $\gamma = 50.4204 \pm 1.45092$
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $149 \leq E$
 $\zeta = -0.0025:0.005:0.3525$ in $[1-\zeta, 1+\zeta, \zeta]$

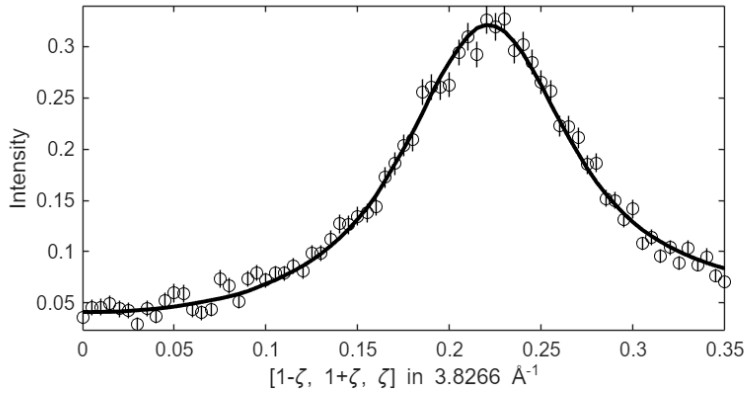


Fe_ei401_align_sym3, w2_400off110.sqw
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $149 \leq E$
 $\zeta = -0.0025:0.005:0.3525$ in $[1-\zeta, 1+\zeta, \zeta]$

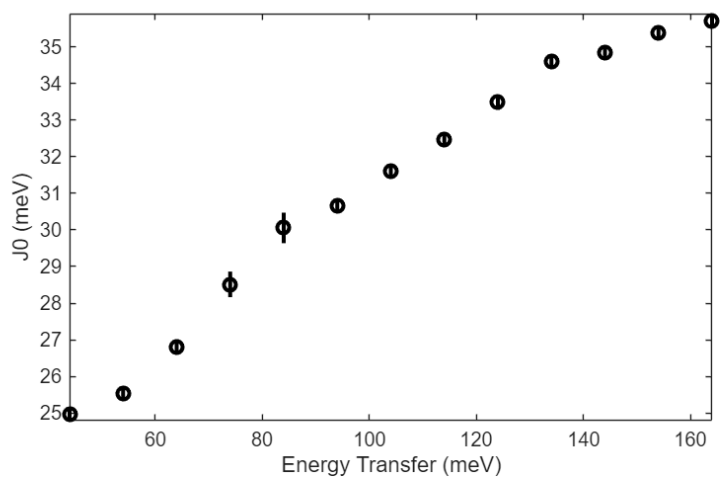
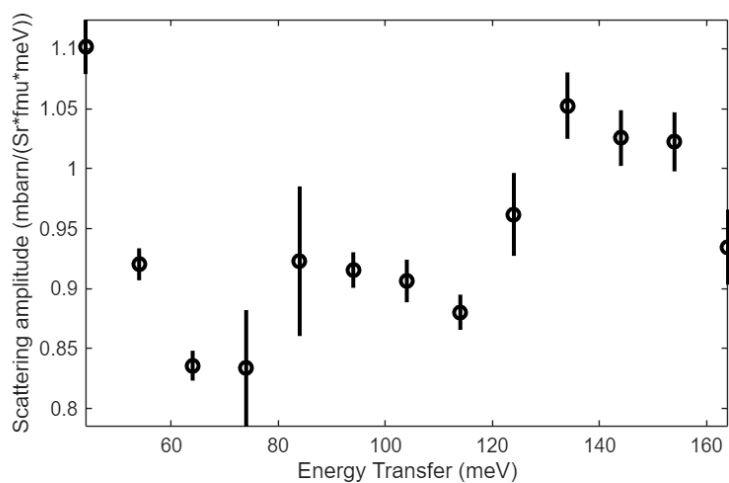
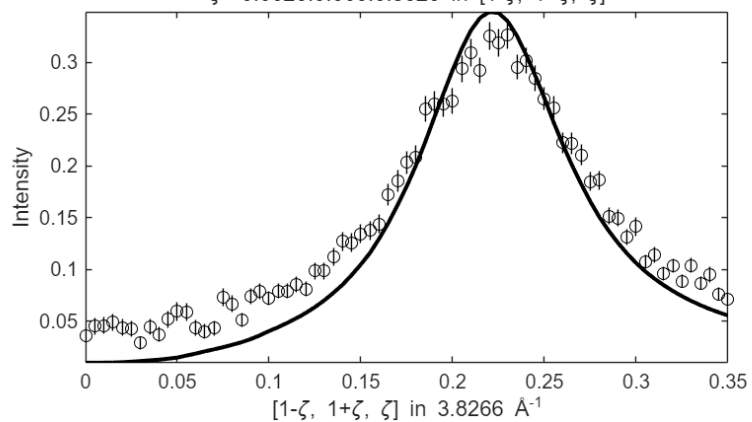


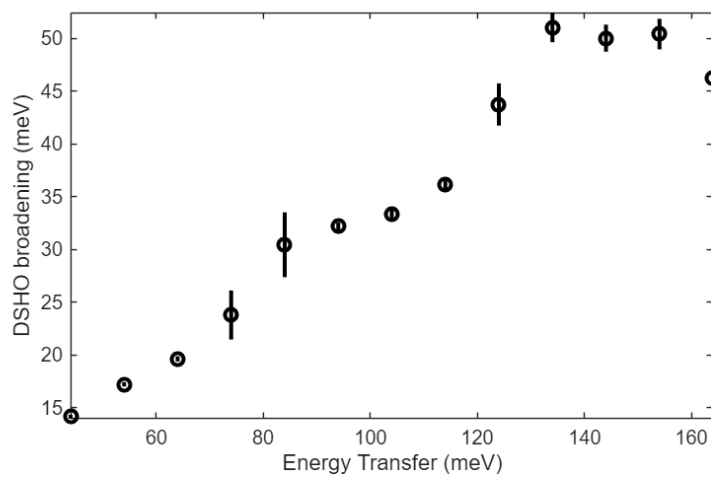
**** En = 164±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw
 $S = 0.93 \pm 0.031$; $J_0 = 36 \pm 0.2$; $\gamma = 46.2872 \pm 1.84331$
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $159 \leq E$
 $\zeta = -0.0025:0.005:0.3525$ in $[1-\zeta, 1+\zeta, \zeta]$



Fe_ei401_align_sym3, w2_400off110.sqw
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $159 \leq E$
 $\zeta = -0.0025:0.005:0.3525$ in $[1-\zeta, 1+\zeta, \zeta]$





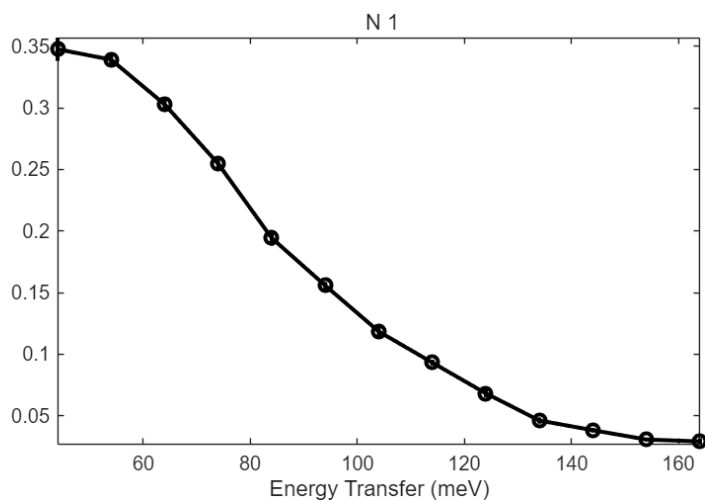
```

cl = GP_FFCutsFits.(selectCuts);
GP_FFCutsFits.(selectCuts) = [cl{:}];

beep
beep
beep

all_fit_par = GP_FFCutsFits.([selectCuts, '_all_fit_par']);
bg_data = extract_bg_par(all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure;

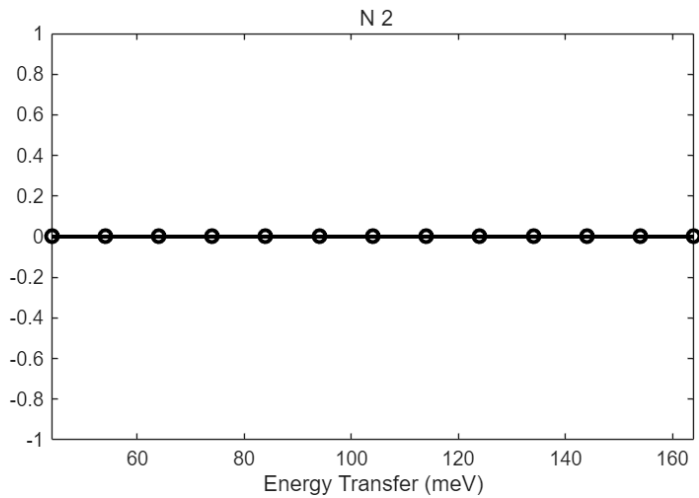
```



```

pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure

```



```
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure;
```

```
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure
```

```
doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\FE\Data\analysis\06_fit_with_AQ0\fit_cutFe_GP.mlx';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_GP_',selectCuts,'.pdf']);
export(thisFile,targ_file);
```

```
Error using export>checkWritePermission (line 251)
Unable to overwrite existing file. Check write permission or whether the file is open in another application.
```

```
Error in export (line 104)
checkWritePermission(results.Destination);
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
    %close all;
%end
```

```
fit_src_struct = GP_FFCutsFits;
save(Cutfitfile,"fit_src_struct",-v7.3')
```